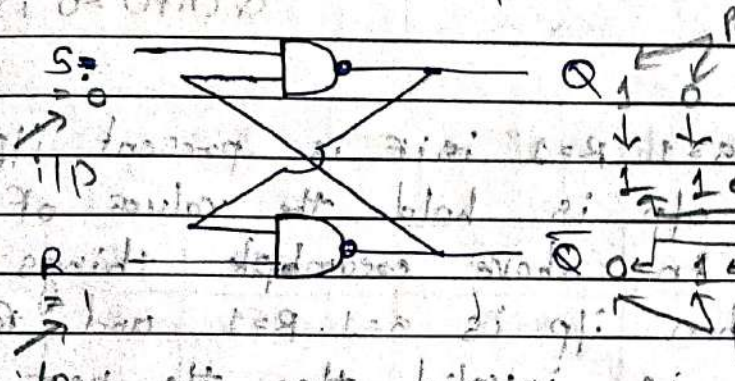
* Sequential Circuits *

- Steady state Response → fixed
- Transient Response → Variable

* S-R latch :-

- S-R latch using NAND Gates
- The o/p of SR latch is depends on three properties ① Present i/p ② Present o/p ③ Next o/p



- So avoid transient o/p that's why first we are dealing with that NAND gate who's i/p is '0'.
- Working :- Next o/p Q(n+1) is function of present i/p and present o/p Q(n)

S	R	Q(n+1)	
0	0	0	invalid state → because whenever we give the i/p S=0, R=0, then in Nex o/p both Q and $\bar{Q}$ are coming same and it is invalid $Q \neq \bar{Q}$.
0	1	1	
1	0	0	
1	1	Q(n)	→ Hold state means जो o/p present o/p में है वही o/p Next o/p me आ रहे है.

S=0, R=1	Q(n) = ?	Q(n+1) = 1
S=1, R=1	Q(n) = 1	Q(n+1) = 1 → Hold
S=1, S=0	Q(n) = 1	Q(n+1) = 0
S=1, R=1	Q(n) = 0	Q(n+1) = 0 → Hold

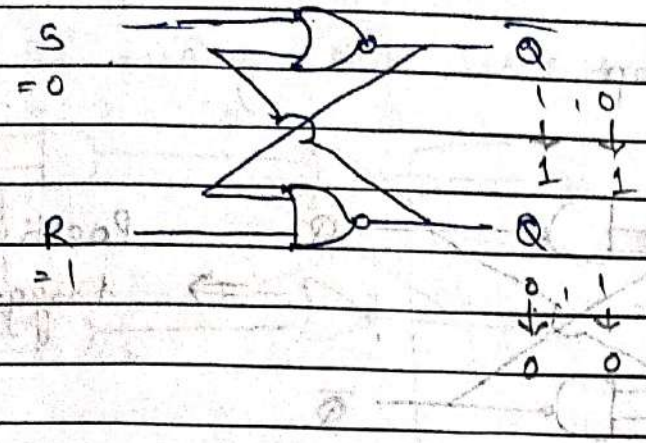
S=0, R=0	Q(n) = 0	Q(n+1) = 1, $\bar{Q}$(n+1) = 1
S=1, R=1	Q(n) = 1, $\bar{Q}$(n) = 1	Q(n+1) = 1, $\bar{Q}$(n+1) = 0 (or) Q(n+1) = 0, $\bar{Q}$(n+1) = 1

- Here when S=1, R=1 if is present i/p then it's next o/p is hold the values of present o/p, but in above example things are change
- Because, when i/p is S=1, R=1 and for that present o/p is invalid then the next o/p is depends on the delays of the two NAND gates present in the SR latch, but next o/p is never hold the values of present o/p in above condition

S=0, R=0	Q(n) = ?	Q(n+1) = 1, $\bar{Q}$(n+1) = 1
S=0, R=1	Q(n) = 1, $\bar{Q}$(n) = 1	Q(n+1) = 1, $\bar{Q}$(n+1) = 0

- S-R latch using NOR Gates :-
- So to avoid transcient o/p that's why first we dealing with that NOR gate who's i/p is '1', because in NOR if one of the i/p is '1' then o/p becomes zero

Q(n+1) → Next o/p



S	R	Q(n+1)
0	0	Q(n) → hold state
0	1	0
1	0	1
1	1	invalid state

S	R	Q(n+1)	$\overline{Q}$(n+1)
0	0	Q(n)	$\overline{Q}$(n)
0	1	0	1
1	0	1	0
1	1	0	0 → invalid state

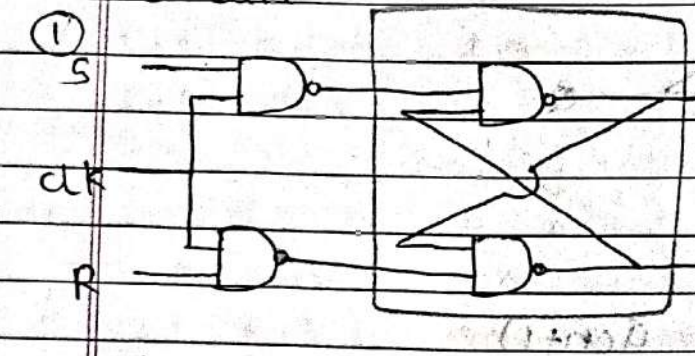
- Note :-
  - Hold state i/p in case of NAND latch or NOR latch will hold the present o/p state in next o/p state only if present o/p state is a valid o/p state.
  - If present o/p state is invalid o/p state then hold state i/p will not be able to hold this invalid o/p state rather in next o/p state →
    → o/p will go to either (1,0) or (0,1).

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* SR Flip-flop :-

- Circuit

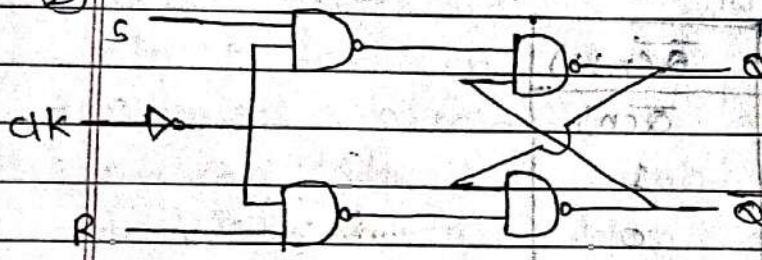
(1)



⇒ Positive level triggered ff

- This flip-flop is enable when clk = 1 and disable when clk = 0 → at clk=0, the values of Q and $\bar{Q}$ hold their value.

(2)



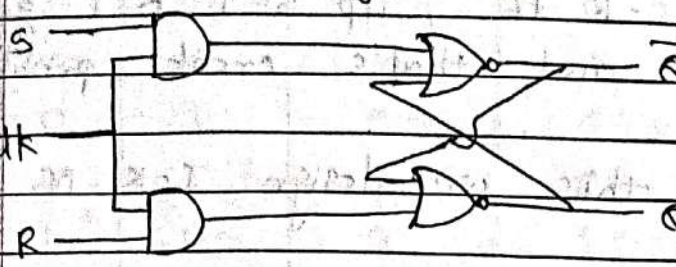
⇒ Negative level triggered ff.

- This flip-flop is enable when clk = 0 and disable when clk = 1

- Working :

clk	S	R	Q(n+1)
Non-active	x	x	Q(n)
Active	0	0	Q(n)
Active	0	1	0
Active	1	0	1
Active	1	1	invalid state

- S-R ff -using NOR latch +clock



⇒ Positive level triggered ff

- Working is same as above table.

- characteristic Table of SR -flip-flop

-When clock Active

S	R	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

$Q(n+1)[S,R,Q(n)] = \Sigma(1,4,5) + d\Sigma(6,7)$

	$\bar{R}\bar{Q}$	$\bar{R}Q$	RQ	$R\bar{Q}$
$\bar{S}$		1		
S	1	1	x	x

$$Q(n+1) = S + \bar{R}Q(n)$$

-characteristic eqn

-Valid when, S,R = 0

- Excitation Table :-

Q(n)	Q(n+1)	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

0 → 0 : S=0, R=0 / S=0, R=1

0 → 1 : S=1, R=0

1 → 0 : S=0, R=1

1 → 1 : S=0, R=0 / S=1, R=0

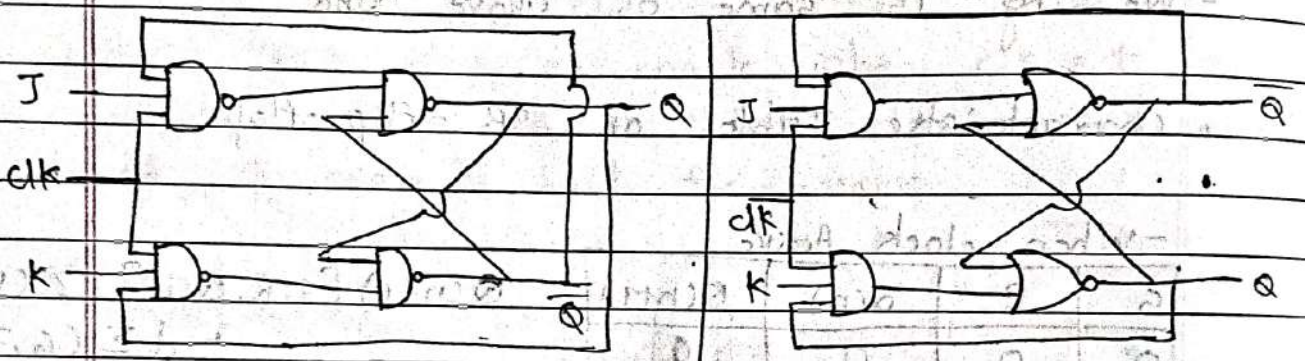
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- Problem with S-R ff

- In case of S-R ff, i/p S=1, R=1 generate invalid o/p state and that's create problem in practical ckts.

↳ To avoid this we design J-K ff

J-K flip flop :-



+ve level triggered ff +ve level triggered ff

- Truth Table :-

CLK	J	K	Q(n+1)	
Non-Active	x	x	Q(n)	
Active	0	0	Q(n)	→ hold state
Active	0	1	0	
Active	1	0	1	
Active	1	1	$\overline{Q(n)}$	→ Toggle state

- characteristic Table :-

Q(n+1) [J, K, Q(n)] = Σ (1, 4, 5, 6)

	$\overline{K}\overline{Q}$	$\overline{K}Q$	KQ	$K\overline{Q}$
$\overline{J}$		1		
J	1	1		1

$$Q(n+1) = J\overline{Q}(n) + \overline{K}Q(n)$$

↳ always valid for any J & K i/p

J	K	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

* Excitation Table :-

Q(n)	Q(n+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

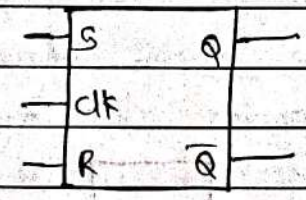
0 → 0 : J=0, K=0 / J=0, K=1

0 → 1 : J=1, K=0 / J=1, K=1

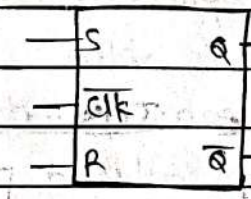
1 → 0 : J=0, K=1 / J=1, K=1

1 → 1 : J=0, K=0 / J=1, K=0

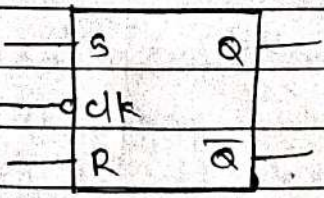
* Different representation of level triggered FF's



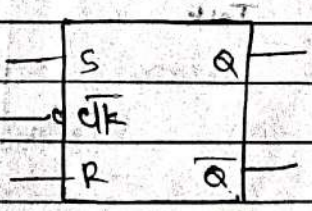
+ve level triggered FF



-ve level triggered FF



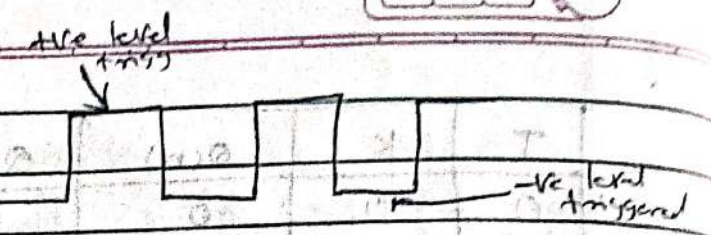
-ve level triggered FF



-ve level triggered FF

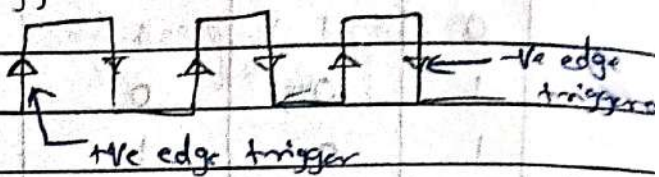


- Level triggered →



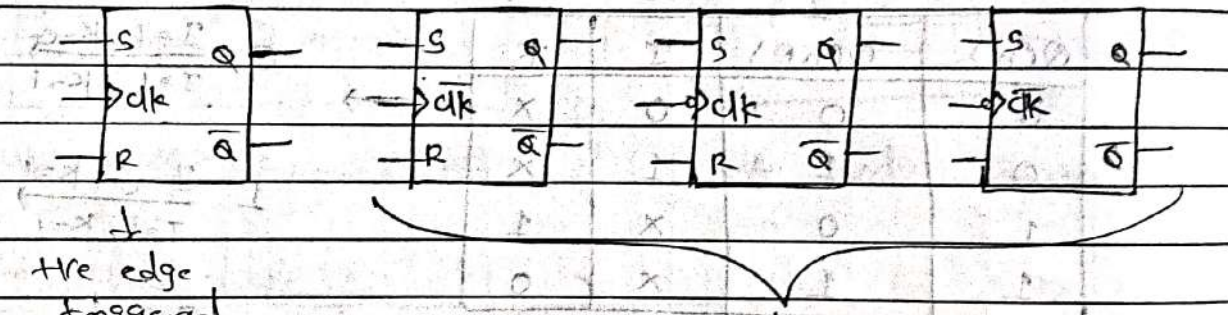
o/p changes while level is triggered it depend on +ve or -ve level trigger

- Edge triggered →



o/p change while edge is triggered it depends on +ve or -ve edge triggered

- Representation of Edge triggered ff

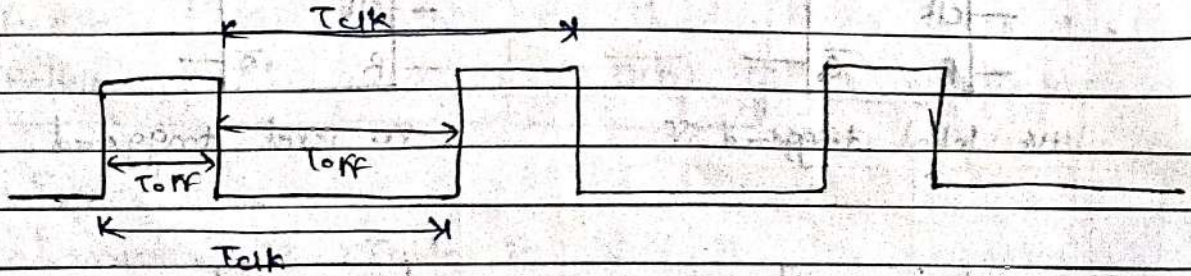


+ve edge triggered

ff is triggered only when clock take transition from (0-1)

-ve edge triggered

ff is triggered only when clock take transition from (1-0)

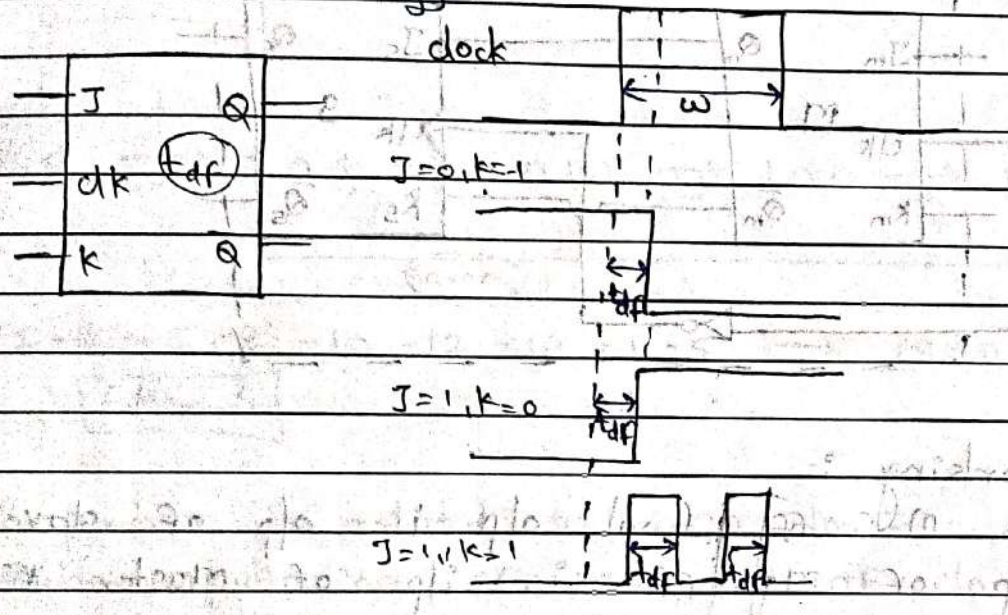


• Notes :-

- In case of level triggered ff o/p may change more than once in single clock cycle.
- In case of edge triggered ff o/p will change only once in a single clock cycle.

* Race around condition :-

+ve level triggered J-K ff



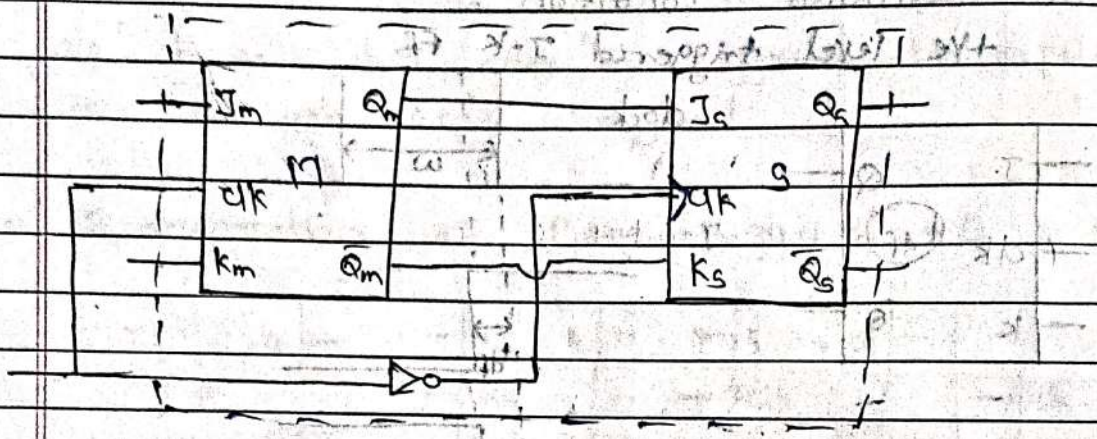
In case of level triggered J-K ff, when J=K=1 i/p is applied and time delay of JK ff is much much smaller than pulse width ($t_{df} << w$) then o/p changes more than once in single clock cycle and that creates problem in designing of ckt. This problem is called as race around condition

• final condition for Race Around condition

- Race around condition happens only if
  - (i) ff is level triggered J-K ff
  - (ii) J=K=1 i/p is applied
  - (iii) pulse width w $t_{df} << w$
- If all condition satisfied then only Race around

- Note :-
- Race around condition never happens in edge triggered J-K ff

- How to avoid race around condition
- Circuit of master-slave ff



- Working :-
- In M-S ff, final o/p is o/p of slave ff
- I/p of M-S ff is i/p of master ff
- In M-S ff, Master ff is level triggered and slave ff is edge triggered

* Counters :-
- What is counter and for what it is used for?
- To count no. of clock pulses applied
- It is used to divide frequency by an integer no. n
- Here if freq reduces the time increases mean pulse width also increases
- Pulse streching

Counter की हर एक state एक clock time period के लिए exist करती है.

- See Pdf → 5.Q → 04, 05

- MOD No. of the counter :-
- Total no. of different states in a counter is called as MOD no. of the counter

eg 0 - 1 - 3 - 5 - 7 → MOD no. = 5
Here highest state is 7 in binary to write 7 we need 3 bits so here also we need 3 ff.
FF = 3

$Q_2 Q_1 Q_0$
0 0 0 $\xrightarrow{1st}$ 0 0 1 $\xrightarrow{2nd}$ 0 1 1 $\xrightarrow{3rd}$ 1 0 1 $\xrightarrow{4th}$ 1 1 1 $\xrightarrow{5th}$ 0 0 0
Same

eg 2 - 4 - 6 - 8 - 10 - 12 - 14 - 1 - 5 → MOD No. = 9
FF = 4

- Properties related to MOD No. :-

① - After application of MOD no. of clock pulses counter reaches to its starting state.
- After integer multiple of MOD no. of clock pulses, counter reaches to its starting state.

② $f_{clk} = f_{in}$ → [MOD no. n] → $\frac{f_{clk}}{n} = f_{out}$

- The output frequency of a counter is equal to the input frequency of counter divided by MOD no.

eg. 0 - 1 - 3 - 2 → MOD no. = 4
FF = 2

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	Q_1	Q_0
0 →	0	0
1 →	0	1
3 →	1	1
2 →	1	0

$Q_0 \rightarrow T_{ON} = 2T_{clk}$, $T_{OFF} = 2T_{clk}$

$T_{Q_0} = T_{ON} + T_{OFF} = 4T_{clk}$

$f_{Q_0} = \frac{1}{T_{Q_0}} = \frac{1}{4T_{clk}} = \frac{1}{4 \times \frac{1}{f_{clk}}} = \frac{f_{clk}}{4}$

$$\boxed{f_{Q_0} = \frac{f_{clk}}{4}}$$

$Q_1 \rightarrow T_{ON} = 2T_{clk}$, $T_{OFF} = 2T_{clk}$

$T_{Q_1} = 4T_{clk}$, $\boxed{f_{Q_1} = \frac{f_{clk}}{4}}$

- Not every output frequency of every flip flop is divided by MOD no. but any one of the flip flop is having output frequency equal to input freqn divided by MOD. no.

eg $0 - 3 - 2 - 5 - 4 - 7 \rightarrow$ MOD no $= 6$

ff = 3

	Q_2	Q_1	Q_0
0 →	0	0	0
3 →	0	1	1
2 →	0	1	0
5 →	1	0	1
4 →	1	0	0
7 →	1	1	1

$Q_0 \rightarrow T_{ON} = 1T_{clk}$, $T_{OFF} = 1T_{clk}$

$T_{Q_0} = 2T_{clk}$, $\boxed{f_{Q_0} = \frac{f_{clk}}{2}}$

$Q_1 \rightarrow$ [0] ['1'] [0 0] → Disjoint waveform

$\boxed{f_{Q_1} = \frac{f_{clk}}{6}}$

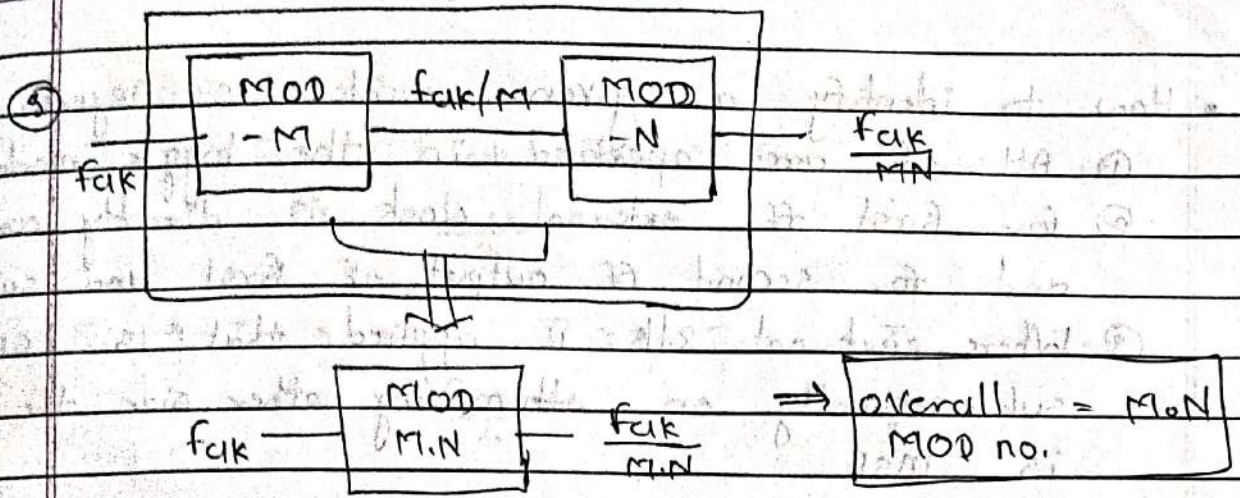
$Q_2 \rightarrow T_{OFF} = 3T_{clk}$, $T_{ON} = 3T_{clk}$

$T_{Q_2} = 6T_{clk}$, $\boxed{f_{Q_1} = \frac{f_{clk}}{6}}$

- In above example two flip flops has o/p freqn equal to i/p freqn divided by MOD no.
- The disjoint waveform is happen because Q_1 doesn't have the waveform in repeated manner

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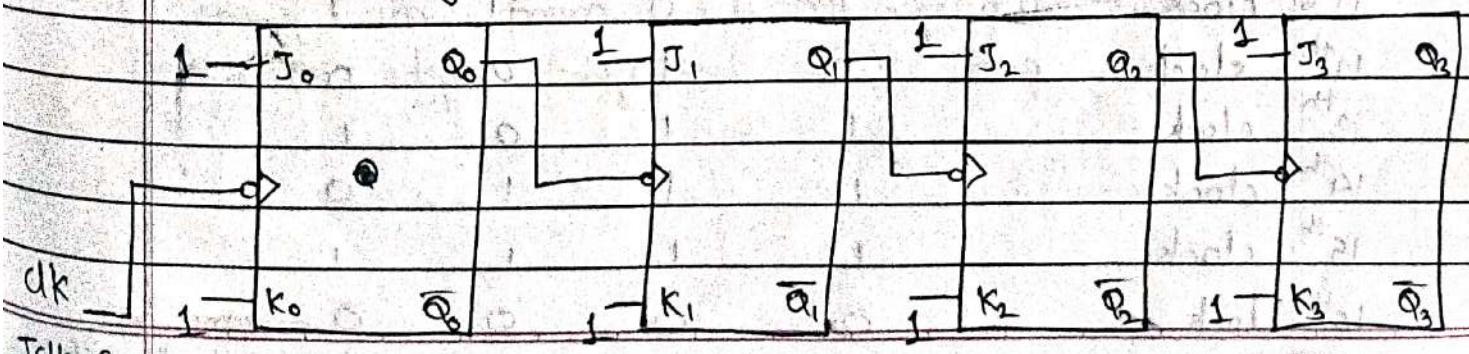
DATE



* Types of counters :-

Asynchronous Counter	Synchronous Counter
- Different ff are driven by different clocks	- All the ffs are driven by same clock
- Fixed sequence are only possible. Sequence → up sequence, → down sequence	- Any sequence is possible
- Delay is more → freq of operation is less i.e Relatively slower	- Delay is less → freq of operation is more i.e. Relatively faster eg. Johnson counter, Ring counter

* 4- bit Asynchronous Counter :-



Tclk, fclk

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- How to identify a given counter is Asynchronous
  ① All ff are operated in the toggle mode
  ② for first ff external clock is directly connect and for second ff output of first and so on
  ③ Where external clk is applied that is LSB a sub-sequently on ~~eth~~ moving other side the last is MSB

- Working :- [4-bit Asynch counter]
  - Q_0 toggles at every clock cycle
  - Q_1 toggles when Q_0 take transition from (1-0
  - Q_2 toggles when Q_1 take transition from (1-0)
  - Q_3 toggles when Q_2 take transition from (1-0)

	Q_3	Q_2	Q_1	Q_0	
Starting state	0	0	0	0	
1st clock	0	0	0	1	
2nd clock	0	0	1	0	
3rd clock	0	0	1	1	
4th clock	0	1	0	0	
5th clock	0	1	0	1	
6th clock	0	1	1	0	
7th clock	0	1	1	1	
8th clock	1	0	0	0	
9th clock	1	0	0	1	
10th clock	1	0	1	0	
11th clock	1	0	1	1	
12th clock	1	1	0	0	
13th clock	1	1	0	1	
14th clock	1	1	1	0	
15th clock	1	1	1	1	
16th clock	0	0	0	0	= same as starting state

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MOD no. = 16

$Q_0 \rightarrow T_{ON} = 1T_{clk}$, $T_{OFF} = 1T_{clk}$, $T_{Q_0} = 2T_{clk}$

$f_{Q_0} = \frac{f_{clk}}{2}$

$Q_1 \rightarrow T_{ON} = 2T_{clk}$, $T_{OFF} = 2T_{clk}$, $T_{Q_1} = 4T_{clk}$

$f_{Q_1} = \frac{f_{clk}}{4}$

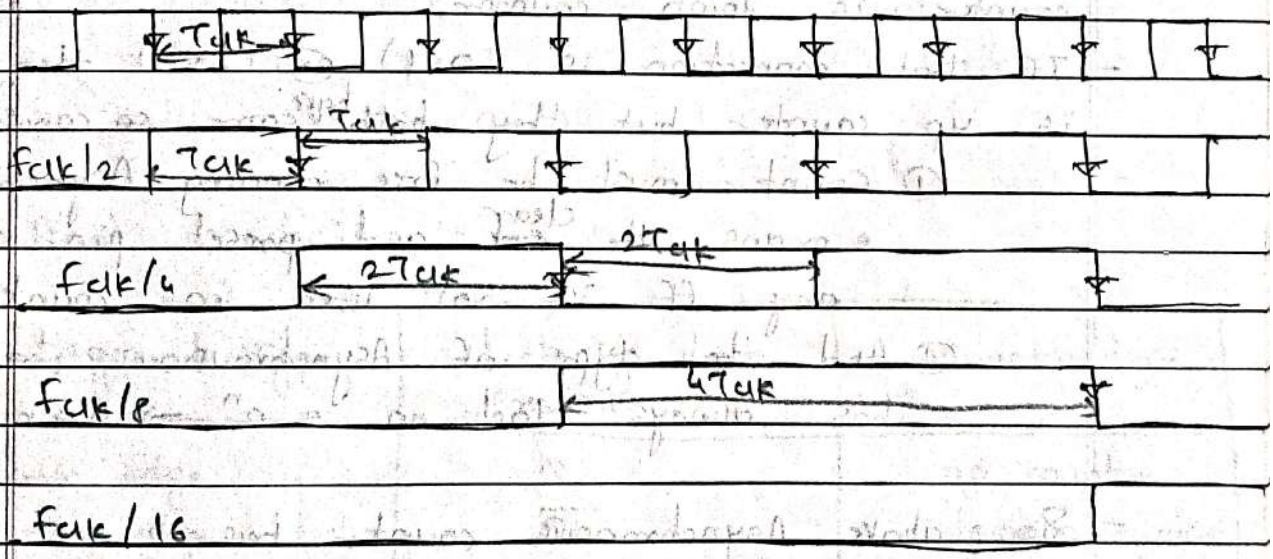
$Q_2 \rightarrow T_{ON} = 4T_{clk}$, $T_{OFF} = 4T_{clk} \rightarrow T_{Q_2} = 8T_{clk}$

$f_{Q_2} = \frac{f_{clk}}{8}$

$Q_3 \rightarrow T_{ON} = 8T_{clk}$, $T_{OFF} = 8T_{clk}$, $T_{Q_3} = 16T_{clk}$

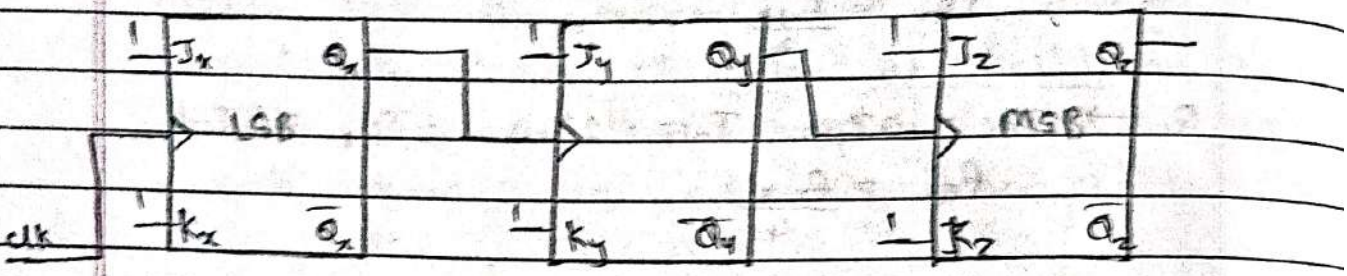
$f_{Q_3} = \frac{f_{clk}}{16}$

- Clock Diagram :-



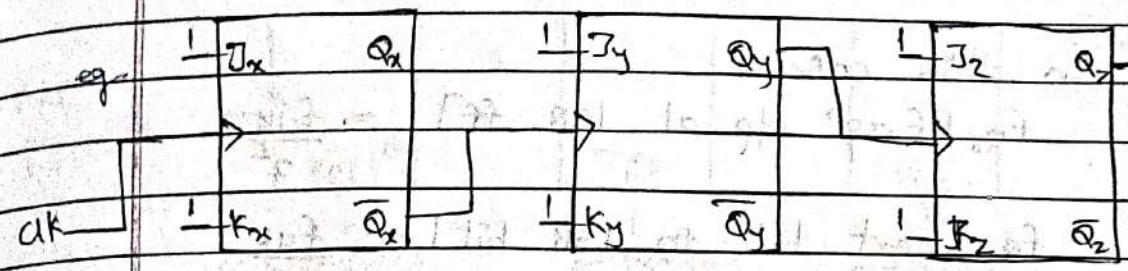
- Here clock is toggle when external clock is goes from from (1-0) because we use negative edge triggered flip flop

- How to Identify LSB bit and type (up or down) of Asynchronous counter.



- external clock is applied at which that flip flop is LSB and moving forward last one is MSB
- Here all positive edge triggered flip flop are used so $Q_x \rightarrow 1$ means (1-1), if flip flop is negative edge triggered and $\overline{Q}_x$ is connected $\overline{Q}_x \rightarrow 0$ (0-0) so when (1-1) or (0-0) is connected then that counter is down counter.
- If that connection is (0-1) & (1-0) then counter is up counter but they both have some so condition
  1. Counter must be free running Asynchronous & means the clear and preset pin of any ff is not used in counter
  2. And that type of Asynchronous ~~des~~ counter has always Mod no. $= 2^n \rightarrow n \rightarrow$ no. of ff

- So above Asynchronous counter has →
  LSB → Q_x
  MSB → Q_z
  - It is down counter
  - Its mod no. is → $2^n = 2^3 = 8$



Mod no. $= 2^3 = 8$

Sequence is

Q_z	Q_y	Q_x
0	0	0
0	0	1
1	1	0
1	1	1
1	0	0
1	0	1
0	1	0
0	1	1
0	0	0

$f_x = f_{clk}/2$
$f_y = f_{clk}/4$
$f_z = f_{clk}/8$

0 - 1 - 6 - 7 - 4 - 5 - 2 - 3

- Final conclusion :-

Triggered Type	check i/p from previous o/p	Counter type
-ve edge	Q	up counter
-ve edge	$\overline{Q}$	down counter
+ve edge	Q	down counter
+ve edge	$\overline{Q}$	up counter

- Note :-

① In case of free running asynchronous counter designed with n-ff → The mod no. is always 2^n

↳ whether counter is connected in up counter mode or down counter or improperly connected

@ And in this case

f_o [Freq^n o/p at LSB ff] = $\frac{f_{clk}}{2^1}$

f_o [next bit to LSB bit] = $\frac{f_{clk}}{2^2}$

⋮

f_o at MSB bit of f_o = $\frac{f_{clk}}{2^n}$

* Clear and Preset input :-

→ CLR (Reset) and PRESET (Set) are pin of flip flop

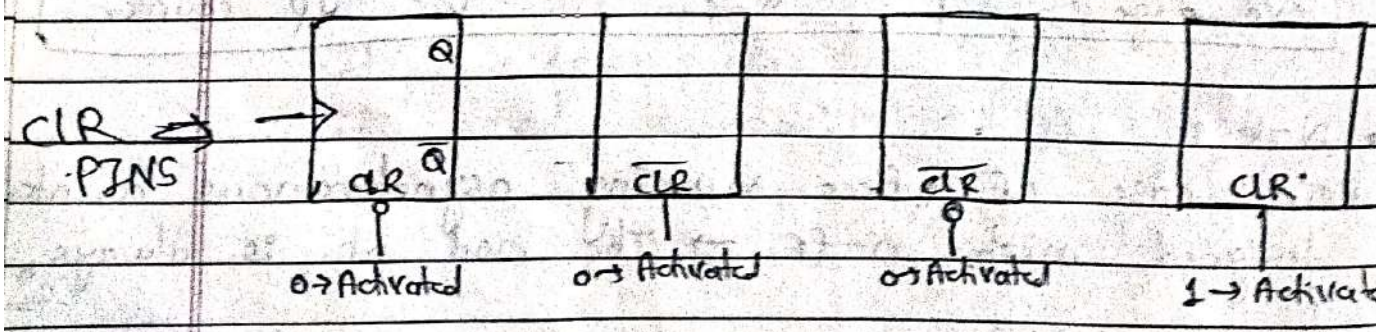
① Clear i/p (or) CLR PIN i/p :-

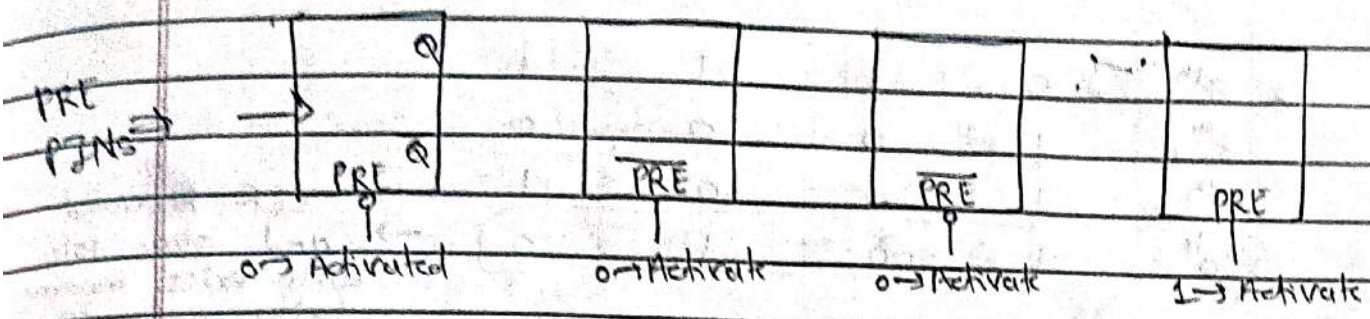
It is by default asynchronous in nature and when activated then irrespective of i/p J and K and clock i/p it resets (clear) the o/p Q.

② ~~Preset~~ Preset i/p (or) PRE PIN i/p :-

It is by default asynchronous in nature and when activated then irrespective of i/ps J & K and clock i/p, it sets the o/p Q

• low Enable and High Enable CLR and PRE i/p:



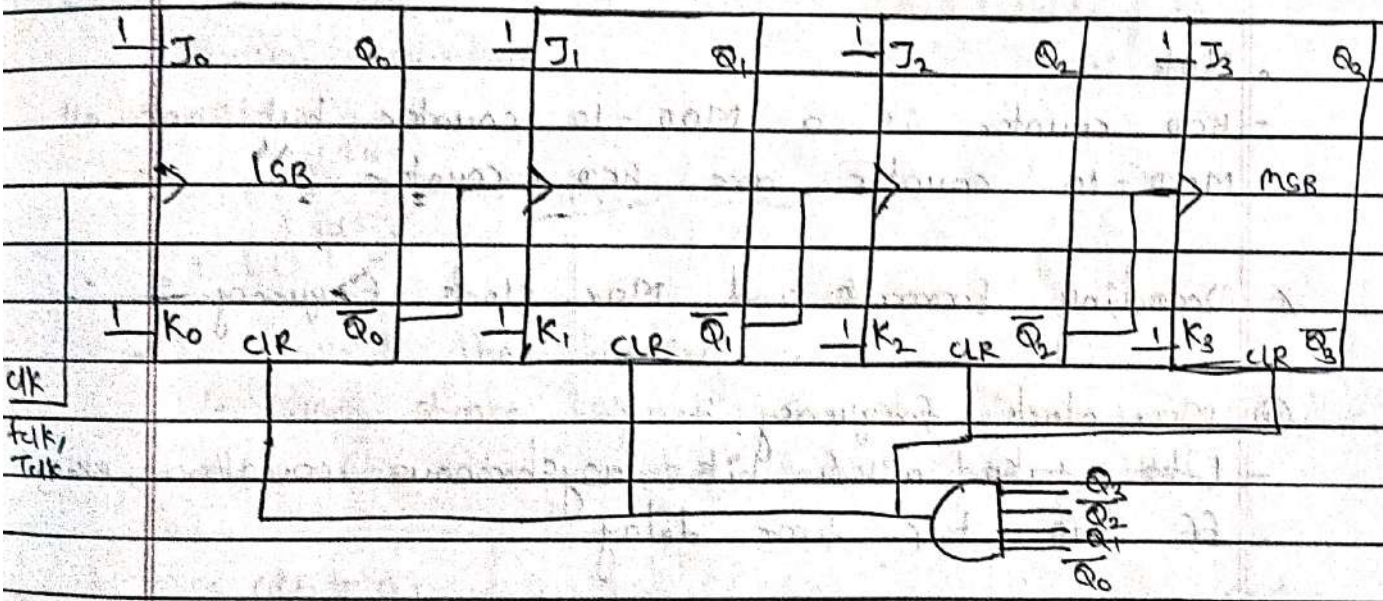


- Designing of Asynchronous Counter with MOD no. other than 2^n.

- BCD Counter Design

BCD counter counts → 0 - 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9

Mod no. = 10 , ff = 4



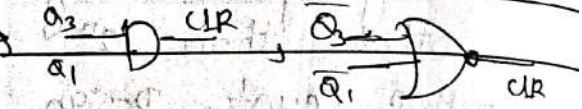
- Table :-

	fclk/10	fclk/10	fclk/10	fclk/2	→ disjoint waveform
CLR	Q_3	Q_2	Q_1	Q_0	
Start	0	0	0	0	
1st	0	0	0	1	
2nd	0	0	1	0	
3rd	0	0	1	1	
4th	0	1	0	0	
5th	0	1	0	1	
6th	0	1	1	0	

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7th	0	1	1	1
8th	1	0	0	0
9th	1	0	0	1
10th	1	0	1	0

→ CLR Activated and state 1010 will exist momentarily and will not be counted in MOD no.

- We can also ~~take~~ take → 

- MOD 12 counter → CLR will be activated at combination $(1010)_2 \rightarrow (12)_{10}$

- Note :-

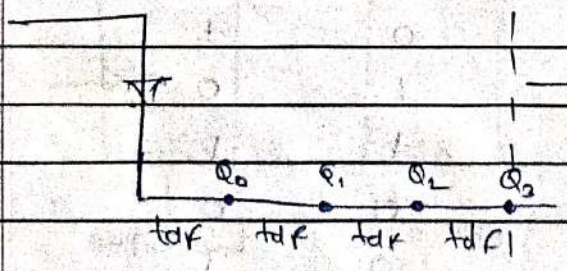
- BCD counter is a MOD-10 counter but not all MOD-10 counter are BCD counter.

* Decoding Error & and Max. clock frequency :-

① Max. clock frequency :-

- Lets take a 4-bit asychronous counter, every ff has t_{df} time delay

- Lets understand worst case delay and max clock freq^n



It just delayed the output

worst case delay $T_d = n t_{df}$

for proper operation $T_{clk} \geq T_d$

$T_{clk} \geq n t_{df} \rightarrow (T_{clk})_{min} = n t_{df}$

$\frac{1}{f_{clk}} \geq n t_{df} \rightarrow f_{clk} \leq \frac{1}{n t_{df}} \rightarrow (f_{clk})_{min} = \frac{1}{n t_{df}}$

② Decoding Error :- [Essential Hazard]

- up counter

1011 → (tdf) 1010 → (tdf) 1000 → (tdf) 1100

1010 and 1000 are transient error or decoding error

- Down counter

0110 → (tdf) 0111 → (tdf) 0101

0111 → transient error

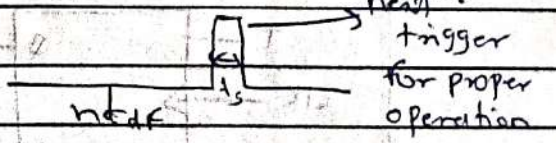
- To avoid the decoding error :-

We use strobe signal so the decoder is active only after the counter output have settled

- See pdf → 07

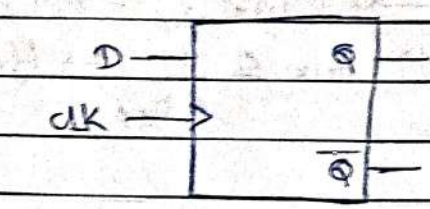
$N_{max} \; (f_{clk})_{max} = \frac{1}{(n t_{df} + t_s)}$

t_s = time delay of strobe signal



* D - FF and T - ff :-

- D - Flip flop :-



$Q(n+1) = D$

- characteristic Table

D	Q(n)	Q(n+1)
0	0	0
0	1	0
1	0	1
1	1	1

$Q(n+1)\,[D, Q(n)] = \Sigma(2,3)$

	$\overline{Q(n)}$	Q(n)
D		
D	1	1

when triggered is applied

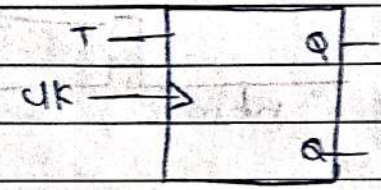
$Q(n+1) = D$

- Excitation Table :-

Q(n)	Q(n+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Note :- level triggered D-FF is called transperent latch

- T - flip flop :-



$T = 1 \rightarrow Q(n+1) = \overline{Q(n)}$

$T = 0 \rightarrow Q(n+1) = Q(n)$

- characteristic Table

T	Q(n)	Q(n+1)
0	0	0
0	1	1
1	0	1
1	1	0

$Q(n+1)\,[T, Q(n)] = \Sigma(1,2)$

$= \overline{T}\,Q(n) + T\,\overline{Q(n)}$

$Q(n+1) = T \oplus Q(n)$

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• Excitation Table :-

Q(n)	Q(n+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

• Toggle mode of operation using J-K flip flop

When → J = 1, K = 1

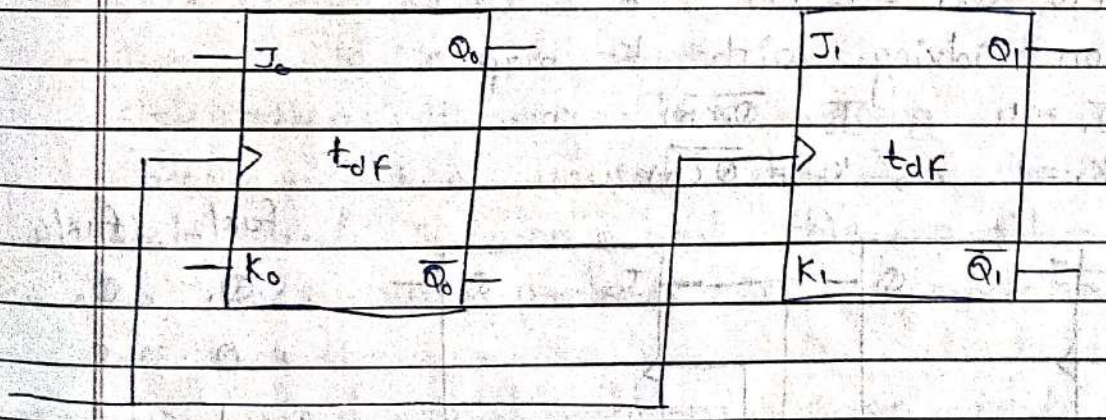
→ J = 1, K = Q(n)

→ J = $\overline{Q(n)}$, K = 1

→ J = $\overline{Q(n)}$, K = Q(n)

* Synchronous Counter :-

- All the flip flop will be driven by same external clock.



- Here both the ff are working parallel so overall delay will be →

Worst case delay $T_d = t_{df}$ for proper operation

$T_{JK} \geq t_{df} \rightarrow (T_{ck})_{min} = t_{df}$

$f_{ck} \leq \frac{1}{t_{df}} \rightarrow (f_{ck})_{max} = \frac{1}{t_{df}}$

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- Decoding Error never occurs in Synchronous counter.

Que : 0 - 3 - 1 - 2

Design above counter using J-K-FF

→

Q_1	Q_0	Q_1^+	Q_0^+	J_1	K_1	J_0	K_0
0	0	1	1	1	x	1	x
0	1	1	0	1	x	x	1
1	0	0	0	x	1	0	x
1	1	0	1	x	1	x	0

- ~~fed~~ fill the J_1, K_1, J_0, K_0 boxes with excitation table

$J_1 [Q_1, Q_0] = \Sigma (0,1) + d\Sigma (2,3)$

$K_1 [Q_1, Q_0] = \Sigma (2,3) + d\Sigma (0,1)$

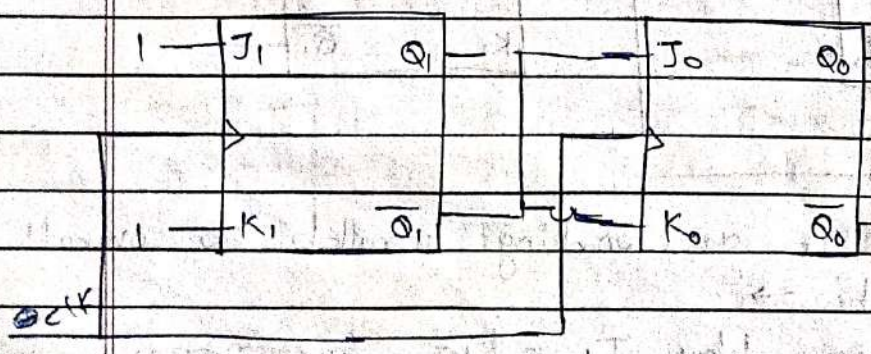
$J_0 [Q_1, Q_0] = \Sigma (0) + d\Sigma (1,3)$

$K_0 [Q_1, Q_0] = \Sigma (1) + d\Sigma (0,2)$

on solving with K- map

$J_1 = 1$, $J_0 = \overline{Q_1(n)}$

$K_1 = 1$, $K_0 = \overline{Q_1(n)}$



fuk/2	fuk/4
Q_1	Q_0
0	0
1	1
0	1
1	0

- If in question given this diagram and we want to find the order then

eqn of J-K ff

$Q_1(n+1) = J_1 \overline{Q_1} + \overline{K_1} Q_1$

$= 1 \cdot \overline{Q_1(n)} + \overline{1} \cdot Q_1(n)$

$\boxed{Q_1(n+1) = \overline{Q_1(n)}}$

$Q_0(n+1) = J_0 \overline{Q_0} + \overline{K_0} Q_0$

$= \overline{Q_1(n)} \cdot \overline{Q_0(n)} + \overline{Q_1(n)} \cdot Q_0(n)$

$= \overline{Q_1(n)} \cdot \overline{Q_0(n)} + Q_1(n) \cdot Q_0(n)$

$\boxed{Q_0(n+1) = Q_1(n) \odot Q_0(n)}$

Q_0	Q_1	
0	0	→ starting state
1	1	Q_1 is toggle of above
1	0	Q_0 is xnor of above
0	1	
0	0	→ like wise

* Lock-Out :-

- Due to noise if counter enters into any unused state then it may lock between one unused state, or two unused states or three unused state and so on and this problem is called as lock-out problem in synchronous counter designing
- To avoid this problem, take next state of unused state must be any of the used state so if counter goes into unused state due to noise then after that it goes into next used state which we are given that place.

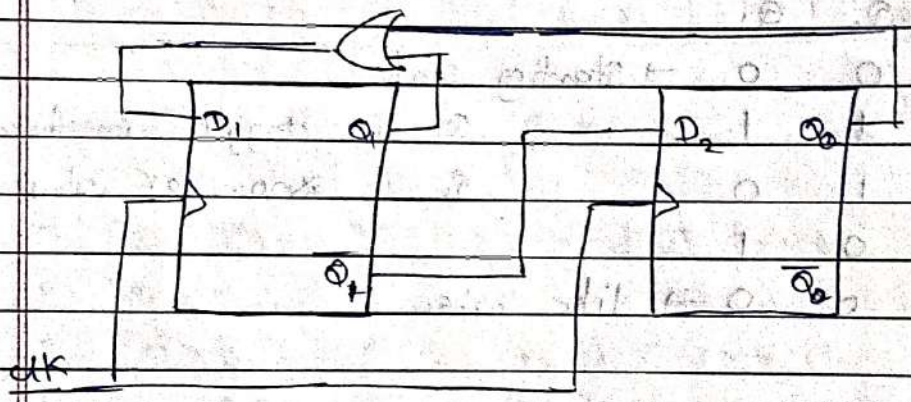
Que : Design a synchronous that counts through stat
0-1-3-2 and saturates at state = (2)

→

	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0
0 → 1	0	0	0	1	0	1
1 → 3	0	1	1	1	1	1
3 → 2	1	1	1	0	1	0
2 → 2	1	0	1	0	1	0

$D_1[Q_1, Q_0] = \Sigma(1,2,3) = (Q_1 + Q_0)$

$D_0[Q_1, Q_0] = \Sigma(0,1) = \bar{Q}_1\bar{Q}_0 + \bar{Q}_1 Q_0 = \bar{Q}_1$



* Synchronous Series carry counter :-
A Synchronous Series carry counter is type of synchronous binary counter in which all flip-flop receive the same clock signal, but the toggle condition of each flip-flop depend on the carry generated by the lower-order flip-flop. The carry is propagated in series, which is why it is called a series carry counter.

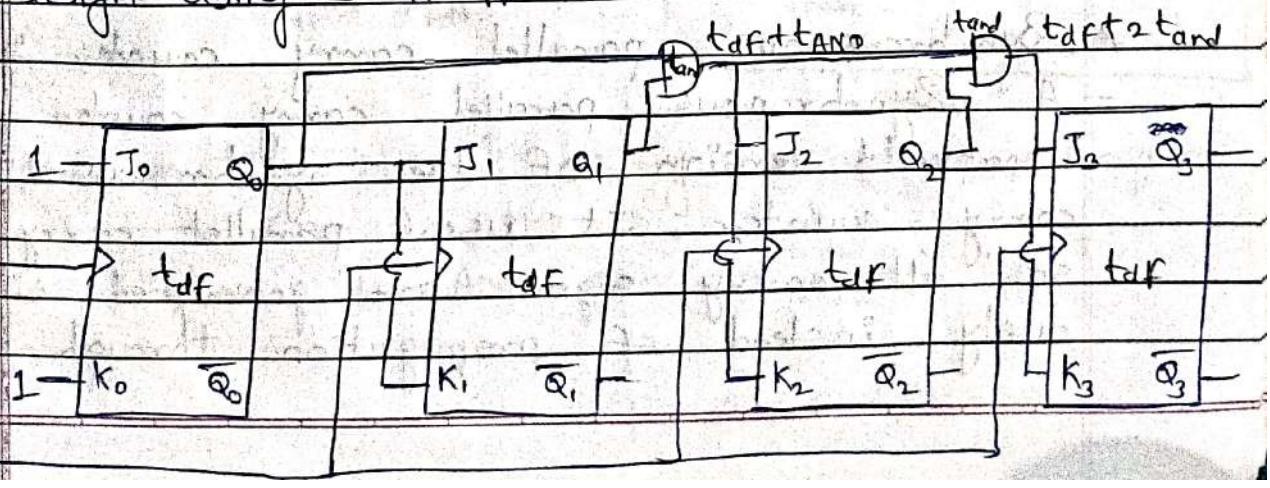
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- 4-bit synchronous series carry counter :-
  - Let's analyze the sequence and design it

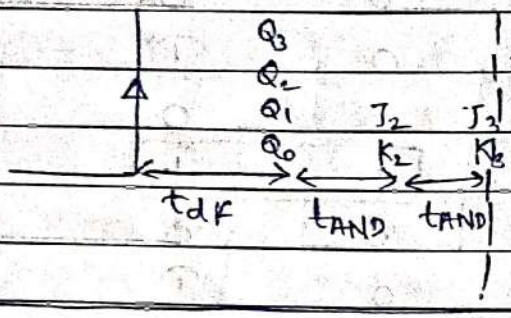
Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	0	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Q_0 toggle after every clock, Q_1 is toggle when Q_0 is 1, Q_2 is toggle when Q_1 & Q_0 both are 1 and Q_3 is toggle when Q_0, Q_1 & Q_2

- Design using J-K ff :-



- No. of AND gate require is :- for n-bit synchronous series carry counter $(n-2)$ [2-i/p AND] gates are required
- Worst case delay and fmax :-



$$\text{Worst case delay} = T_d = t_{dF} + (n-2)\, t_{AND}$$

- for proper operation

$$T_{clk} \geq t_{dF} + (n-2)\, t_{AND}$$

$$(T_{clk})_{min} = t_{dF} + (n-2)\, t_{AND}$$

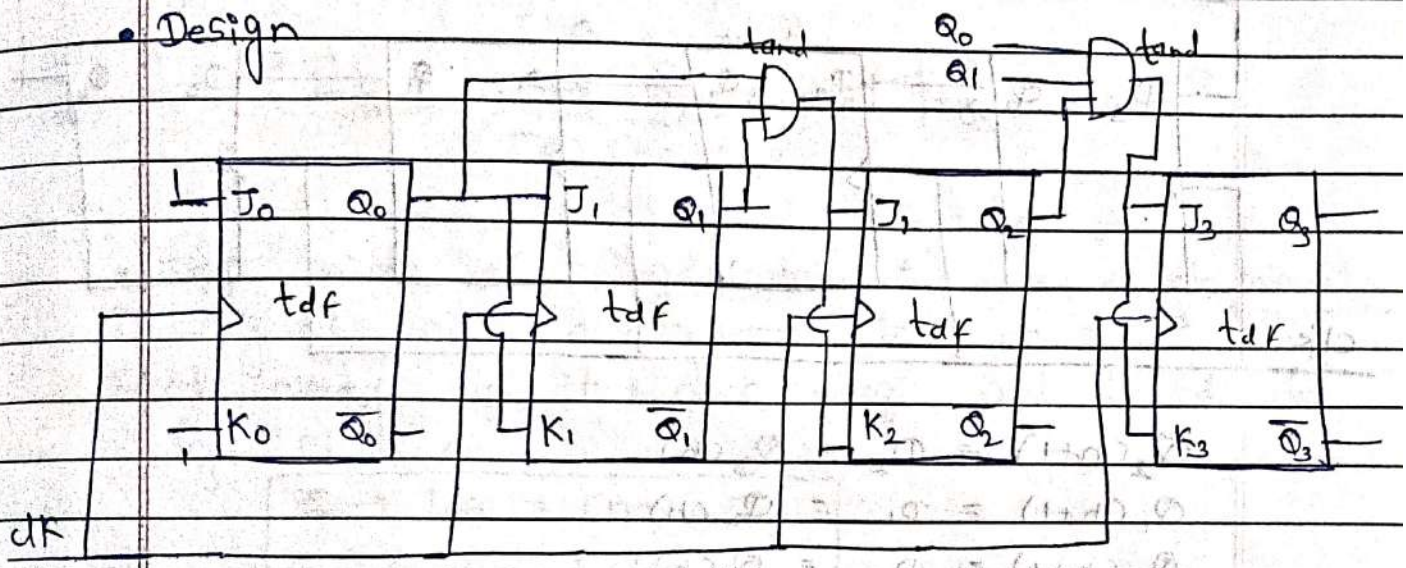
$$f_{clk} \leq \frac{1}{t_{dF} + (n-2)\, t_{AND}}$$

$$(f_{clk})_{max} = \frac{1}{t_{dF} + (n-2)\, t_{AND}}$$

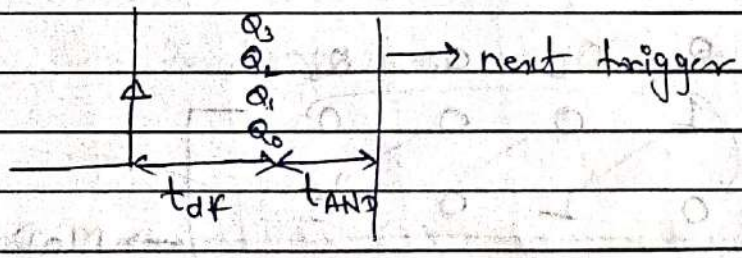
* Synchronous paralled carry counter :-
— A Synchronous paralled carry counter is an improved version of the synchronous series carry counter. It uses paralled carry generator so all carry signal are generated simultaneously instead of propagation through a chain

of AND gates. This significantly reduces propagation delay and makes it suitable for high-speed counters.

- Design



- Worst case delay and f_{max} :



for proper operation

$T_{clk} \geq t_{df} + t_{AND} \rightarrow (T_{clk})_{min} = t_{df} + t_{AND}$

$f_{clk} \leq \frac{1}{t_{df} + t_{AND}} \rightarrow (f_{clk})_{max} = \frac{1}{t_{df} + t_{AND}}$

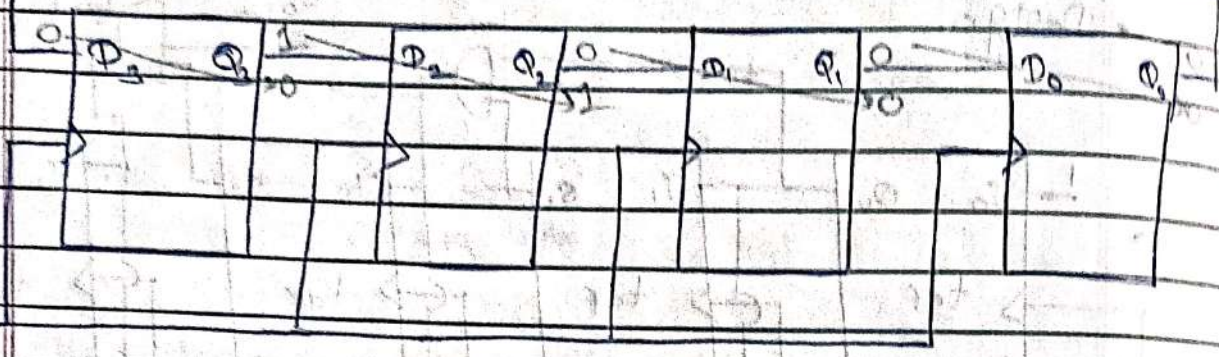
- for n-bit parallel carry counter we require
  - $\rightarrow (n-2)$ AND
  - Highest i/p AND gate $\rightarrow (n-1)$ i/p

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* Ring Counter :-

- 4-bit ring counter

Circuit Design



$Q_3(n+1) = D_3 = Q_3(n)$

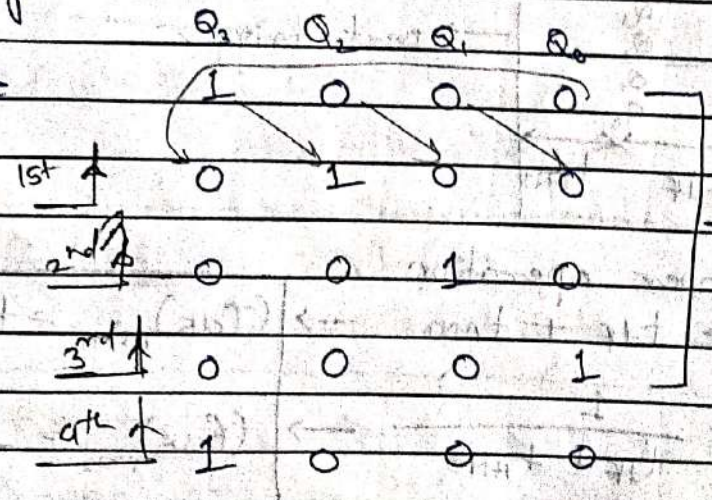
$Q_2(n+1) = D_2 = Q_3(n)$

$Q_1(n+1) = D_1 = Q_2(n)$

$Q_0(n+1) = D_0 = Q_1(n)$

$Q_3(n+1) = D_3 = Q_0(n)$

- Working :-



$T_{ON} = 3\ T_{clk}$

$T_{OFF} = 1\ T_{clk}$

$T_0 = 4\ T_{clk}$

$f_0 = \frac{f_{clk}}{4}$

1 0 1 0] → MOD No → 2

0 1 0 1

1 0 1 0

- Original ~~Johnson~~ Ring counter has starting state and in that starting state all zeros are at one side and 1's at one side

eg 1 0 0 0 → original ring counter
Mod no → 4

1 0 1 0 → Not original ring counter
Mod no → 2

- Imp points of Ring counter :- only original ring counter

① Designed with n-bits → MOD no. = n

② At every o/p i/p clock freq will be divided by MOD no. n

Q → $T_{off} = (n-1)\ T_{clk}$
$T_{on} = 1\ T_{clk}$
$T_Q = n\ T_{clk}$
$f_Q = \frac{f_{clk}}{n}$

% duty cycle at every o/p = $\frac{T_{ON}}{T} \times 100\%$

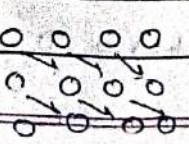
$= \frac{1\ T_{clk}}{n\ T_{clk}} \times 100\%$

% Duty cycle = $\frac{1}{n} \times 100\%$

③ Unused states = $(2^n - n)$

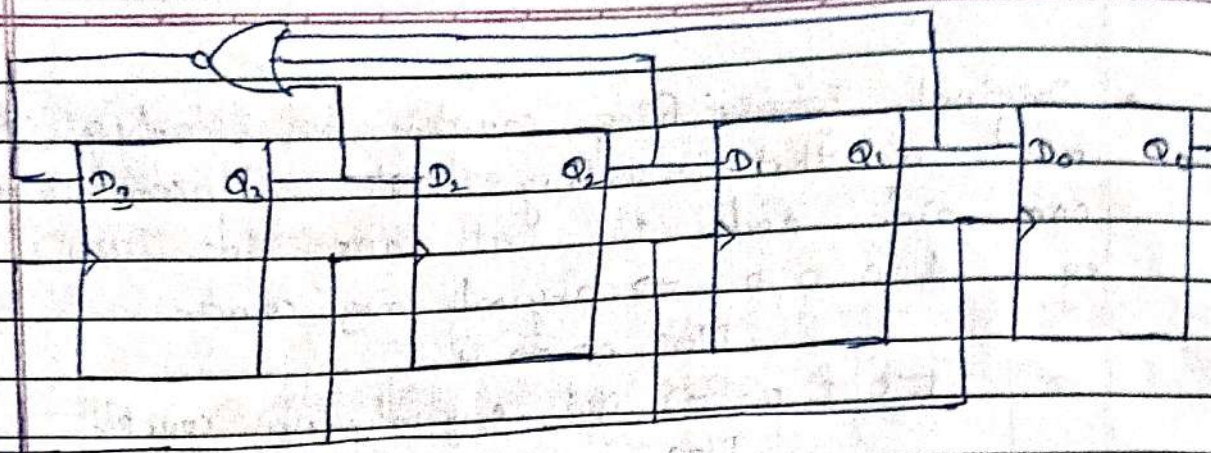
* Self starting Ring counter :-

- In ring counter if we apply first state 0000 then after apply clk that ring counter has stuck and has only 0000 because



like that, to solve this problem we design self starting ring counter

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ok

$Q_2(n+1) = D_2 = Q_3(n)$

$Q_1(n+1) = D_1 = Q_2(n)$

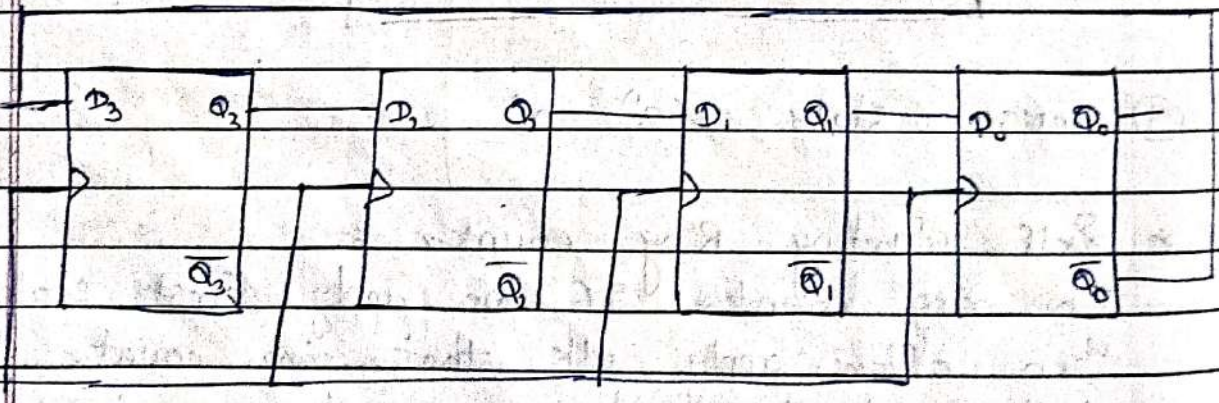
$Q_0(n+1) = D_0 = Q_1(n)$

$Q_3(n+1) = D_3 = \overline{Q_3(n) + Q_2(n) + Q_1(n)}$

Q_3	Q_2	Q_1	Q_0	
0	0	0	0	→ transient state
1	0	0	0	→ MOD no. → 4
0	1	0	0	
0	0	1	0	
0	0	0	1	
1	0	0	0	

* Johnson Counter (or) Twisted Ring Counter :

• 4-bit Johnson Counter :-



$Q_2(n+1) = D_2 = Q_3(n)$ $\quad Q_3(n+1) = D_3 = \overline{Q_0(n)}$

$Q_1(n+1) = D_1 = Q_2(n)$

$Q_0(n+1) = D_0 = Q_1(n)$

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- Working :-

Q_3	Q_2	Q_1	Q_0	
0	0	0	0	
1	0	0	0	compliment
1	1	0	0	
1	1	1	0	→ Mod no. → 8
1	1	1	1	
0	1	1	1	
0	0	1	1	
0	0	0	1	
0	0	0	0	

- To find original Johnson counter same method as ring counter

- Imp points of Johnson counter :- for original Johnson counter

① for n-bit Johnson counter → **Mod no. = 2n**

② At every ff o/p i/p clock freq^n is divided by Mod no.

at every o/p Q $T_{ON} = n\,T_{clk}$

$T_{off} = n\,T_{clk}$

$T_Q = 2n\,T_{clk} = T_{ON} + T_{off}$

$f_Q = \frac{f_{clk}}{2n}$

% duty cycle = 50%

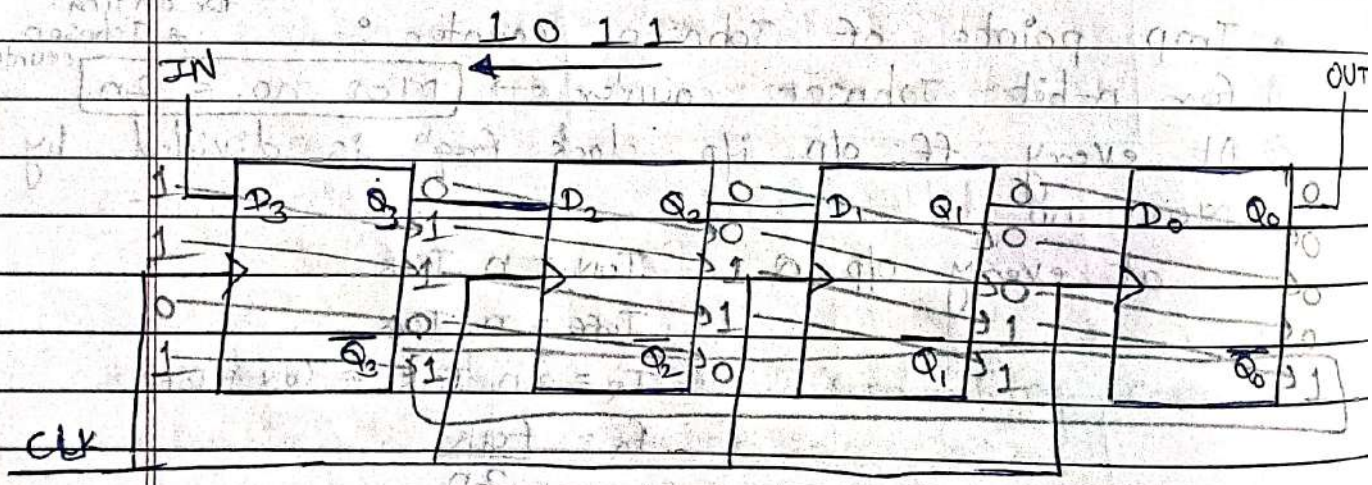
③ Unused state = $2^n - 2n$

* Shift Register :-

- A shift Register is digital circuit made of chain of flip-flops (usually D flip-flops) that stores and moved binary data bit by bit, or position at a time, on each clock pulse.
- Stores information (data - in form of bits)

- Types :-
  1. SISO [Serial IN - Serial OUT]
  2. SIPO [Serial IN - Parallel OUT]
  3. PIPO [Parallel IN - Parallel OUT]
  4. PISO [Parallel IN - Serial OUT]

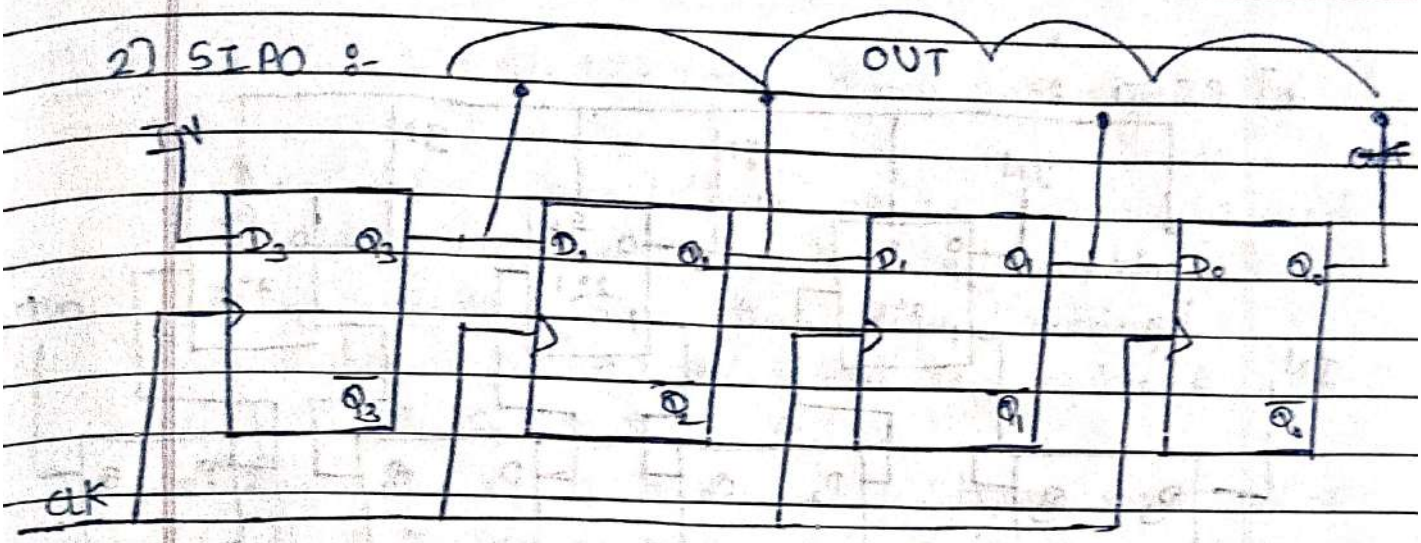
1] SISO :-



- To store n-bit data serially we require n-clocks
- To access n-bit data serially we require (n-1) clocks [(n-1) T_CLK]
- Because the output bit of last flip flop is available coz just connect that output of Q_0 to OUT through wire.

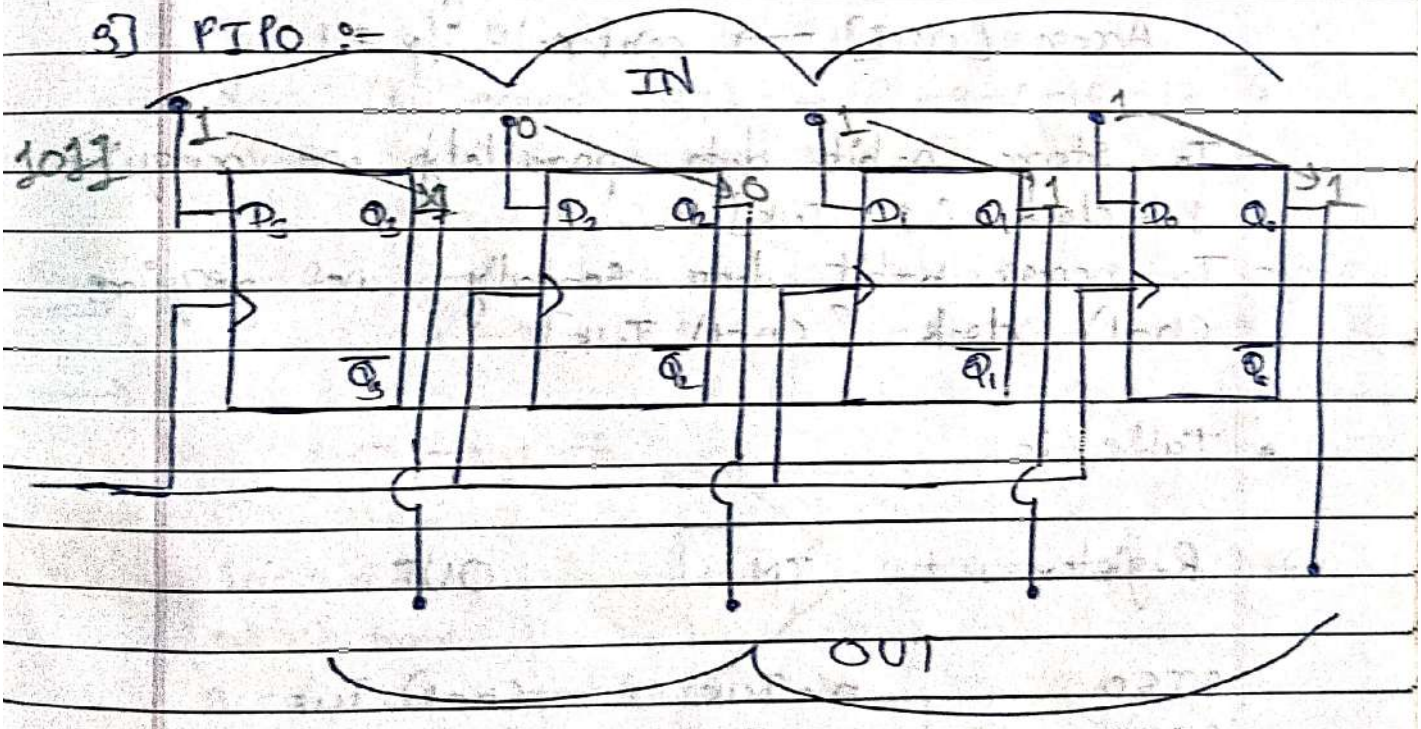
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2] SIPO :-



- To store n-bit serially, we require n-clocks [n-Clk]
- To access n-bit parallelly, we require 0-clocks [0 - T_{clk}]

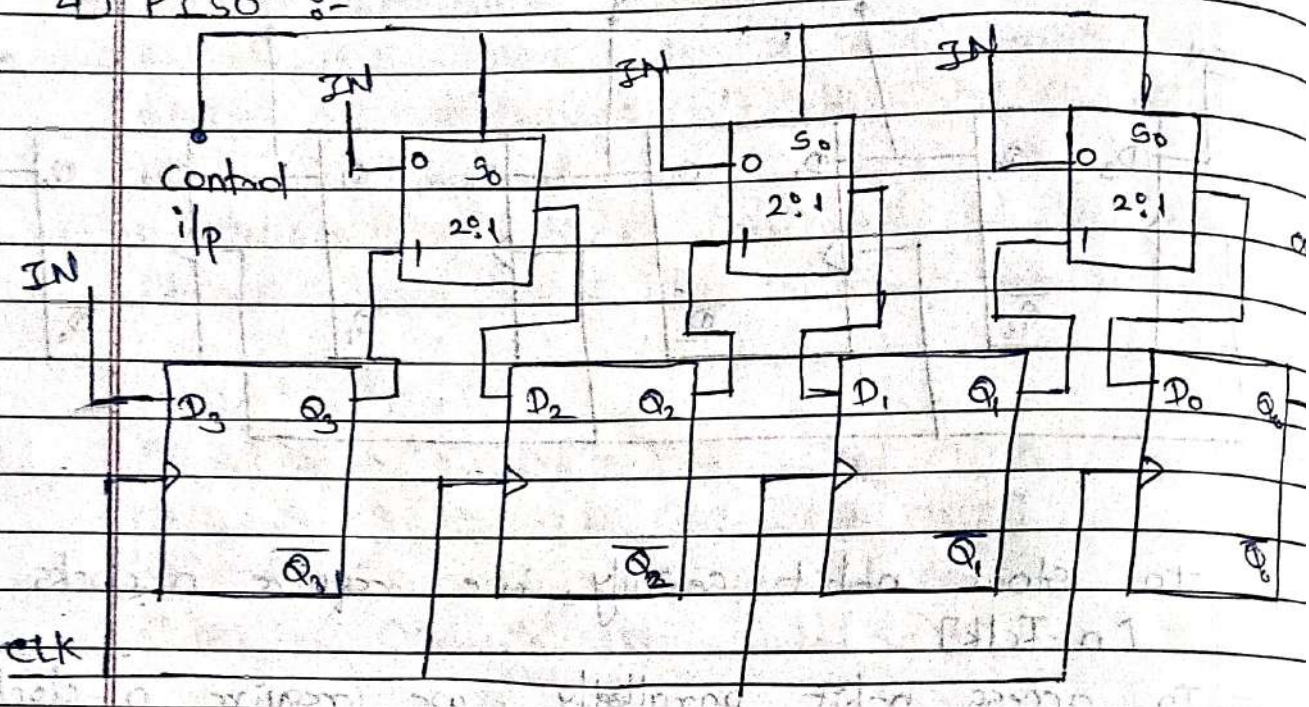
3] PIPO :-



- To store n-bit data parallely we require 1 clock [1 T_{clk}]
- To access n-bit data parallely we require 0-clock [0 T_{clk}]

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4] PISO :-



Store [IN] ⟶ Control i/p = 0
Access [OUT] ⟶ control i/p ≥ 1

- To store n-bit data parallely we require 1 - clock [1 - Tclk]
- To access n-bit data serially we require (n-1) clock [(n-1) Tclk]

- Table :-

Register	IN	OUT
SISO	n - Tclk	(n-1) Tclk
SIPO	n - Tclk	0 Tclk
PIPO	1 - Tclk	(n-1) Tclk
PISO	1 - Tclk	0 Tclk

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Que: 0-1-2-0-1-3-0-1-4. Design th counter

→

Here 1's are three after one 0, 3, and 4 be values are next step so in counter which is put so to solve this type problem

Highest No. → 4 → we require 3-ff

	Q_4	Q_3	Q_2	Q_1	Q_0
[0]	0	0	0	0	0
[1]	0	0	0	0	1
[2]	0	0	0	1	0
[8]	0	1	0	0	0
[9]	0	1	0	0	1
[3]	0	0	0	1	1
[16]	1	0	0	0	0
[17]	1	0	0	0	1
[4]	0	0	1	0	0

Highest most no. of digits are 18 → 3 times

To get 3 different combinations we require two ff

Now it becomes 0-1-2-8-9-3-16-17-4

Now design a counter

Que: 1-2-3-2-3-4-3-2-4-4-2-3-3-4-5-5-3-4-2-2-1-4-3-2-1-2-3-4

→

5 → 3 ff

1 - 3 times

2 - 7 time

3 - 9 times → 4 ff

4 - 7 times

5 - 2 times

3 + 4 = total 7 ff

* Conversion of one ff into another ff

• Procedure :

- Write down the characteristic table of desired ff on i/p side
- Excitation table of available ff on the o/p s
- Now solve the truth table and implement it

• JK to D

JK → Available ff

D → Desired ff

D	Q(n)	Q(n+1)	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

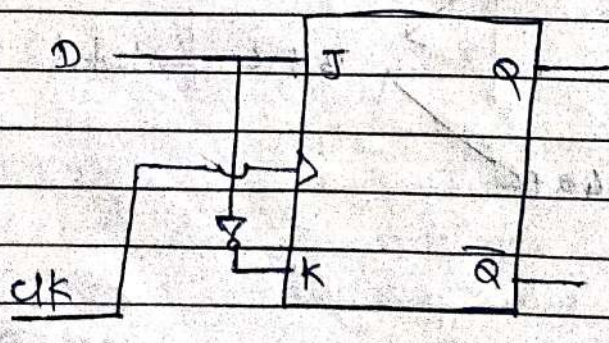
$I[D, Q(n)] = \Sigma(2) + d\Sigma(1,3)$

$K[D, Q(n)] = \Sigma(1) + d\Sigma(0,2)$

on solving using K-map

$J = D$

$K = \bar{D}$



$Q(n+1) = J\bar{Q}(n) + \bar{K}Q$

$= D\bar{Q}(n) + \bar{D}Q$

$= D[1]$

$= $

$Q(n+1) = D$

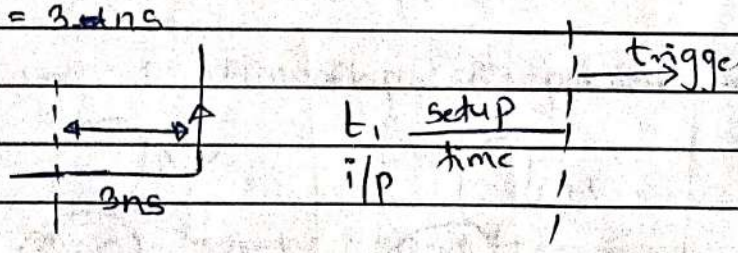
* All conversion Result :-

SR - JK	⟶	$S = J\bar{Q}$, $R = KQ$
SR - D	⟶	$S = D$, $R = \bar{D}$
SR - T	⟶	$S = T\bar{Q}$, $R = TQ$
JK - SR	⟶	$J = S$, $K = R$
JK - D	⟶	$J = D$, $K = \bar{D}$
JK - T	⟶	$J = T$, $K = T$
D - JK	⟶	$D = J\bar{Q}(n) + \bar{K}Q(n)$
D - T	⟶	$D = T \oplus Q(n)$
T - JK	⟶	$T = J\bar{Q}(n) + KQ(n)$
T - SR	⟶	$T = S\bar{Q}(n) + RQ(n)$
T - D	⟶	$T = D \oplus Q(n)$

* Preset time and Hold time :-

① Preset time (setup time) :-

- $t_{setup} = 3ns$

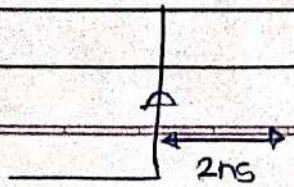


- It is the minimum time before arrival of clock before which i/p must be maintained at proper level for o/p to change properly.

② Hold time :-

- It is the minimum time after arrival of clock for which i/p must be maintained at the same level for proper o/p change

$t_H = 2ns$



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